

Product Requirements Document

HuaweiCloud Agentic Tools — AI-Powered Landing Zone Deployment

Field	Detail
Version	v0.2
Status	In Review
Owner	PM-Lead
Last Updated	2026-05-24
Repo	https://github.com/liubiedi/huaweicloud-agentic-tools

Version History

Version	Date	Author	Summary
v0.1	2026-05-18	PM-Lead	Initial draft — agentic runtime MVP scope
v0.2	2026-05-24	PM-Lead + Claude	Revised workflow: Excel LLD → email-based gap-fill → terraform apply. Full service inventory added. Workflow diagram embedded.

1. Executive Summary

This product automates Huawei Cloud Landing Zone deployment end-to-end — from a structured design document filled by a cloud engineer, through AI-assisted validation, to production infrastructure created via Terraform. The target outcome is a **fully deployed baseline Landing Zone in under 30 minutes with zero manual console steps.**

The system is designed in two layers that can be built and shipped independently:

Layer 1 — Terraform Library (Months 1–2): Thirteen reusable, validated Terraform modules covering all nine Huawei Cloud CAF governance domains. Deterministic. No AI required to run.

Layer 2 — Agentic Workflow (Month 3+): An AI agent that reads a structured Excel LLD, identifies gaps via email, validates against OPA and CAF policies, generates Terraform configuration, and triggers deployment through a CI/CD pipeline. The Terraform modules from Layer 1 are the building blocks.

This PRD primarily covers Layer 2 — the agentic workflow — which is the novel component. Layer 1 is specified in the repository's module documentation.

2. Problem Statement

Building a Huawei Cloud Landing Zone today takes **3–5 days of manual console work** per deployment:

- No reusable IaC exists — every engagement starts from zero
- Knowledge is tribal — two engineers know how, they become the bottleneck
- Deployments are snowflakes — no two are consistent
- Three months post-deployment, no one can tell what's actually running vs what should be
- Onboarding new cloud engineers requires significant mentoring time

The underlying cause is not a skills gap — it is the absence of a **repeatable, code-driven, validated deployment system**.

3. Goals and Success Metrics

Primary goal

Deploy a baseline Huawei Cloud Landing Zone in **under 30 minutes** with **zero manual console steps**, triggered by a single structured document.

Success metrics

Metric	Target	Measurement
Wall-clock deployment time	< 30 min (t0 to completion)	Timestamped agent logs
Manual console steps	0	CTS audit — all API calls must originate from agent IAM agency
LLD gap-fill rounds	≤ 2 email rounds	Agent run logs
Gate pass rate (first attempt)	≥ 80% by Week 6	CI pipeline metrics
Agent self-correction rate	≥ 60% of gate failures	Agent transcript analysis
State recovery time (from OBS versioning)	< 5 min	Tested in chaos drill
Team: new LZ deployment time vs baseline	90% reduction	Before/after comparison

4. Scope

In scope — MVP (Months 1–3)

- All 13 Terraform modules covering 9 Huawei CAF governance domains
- Excel LLD template covering all module input parameters
- Agent workflow: LLD parse → email gap-fill → OPA validation → CAF advisory → tfvars generation → terraform plan → email approval → CI/CD apply → post-apply verification
- Gmail-based communication channel (MVP only — production will use WeCom/Lark)
- Huawei CodeArts Pipeline or GitHub Actions as CI/CD runner
- OIDC-based keyless authentication to Huawei Cloud IAM
- OBS S3-compatible state backend with versioning + KMS encryption
- OPA policy library: public OBS deny, SCP v5 syntax enforcement, required tags, region allowlist, inline resource block deny

Out of scope — MVP

- WeCom/Lark bot integration (production phase)
- Web UI / dashboard

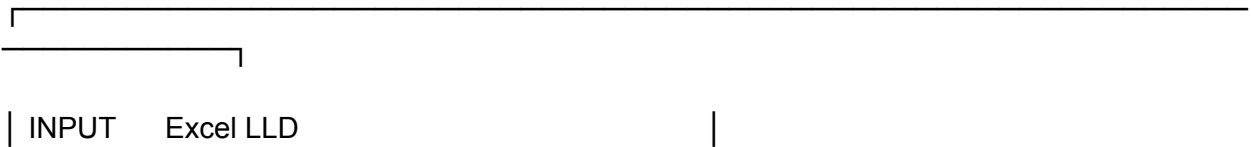
- Multi-region LZ (single-region baseline only)
- Account vending automation beyond the 6 standard LZ accounts
- Advanced security services: DBSS, DSC, AAD (partial Terraform coverage — console-managed)
- Drift remediation agent
- Cost optimization recommendations
- Cross-org federation

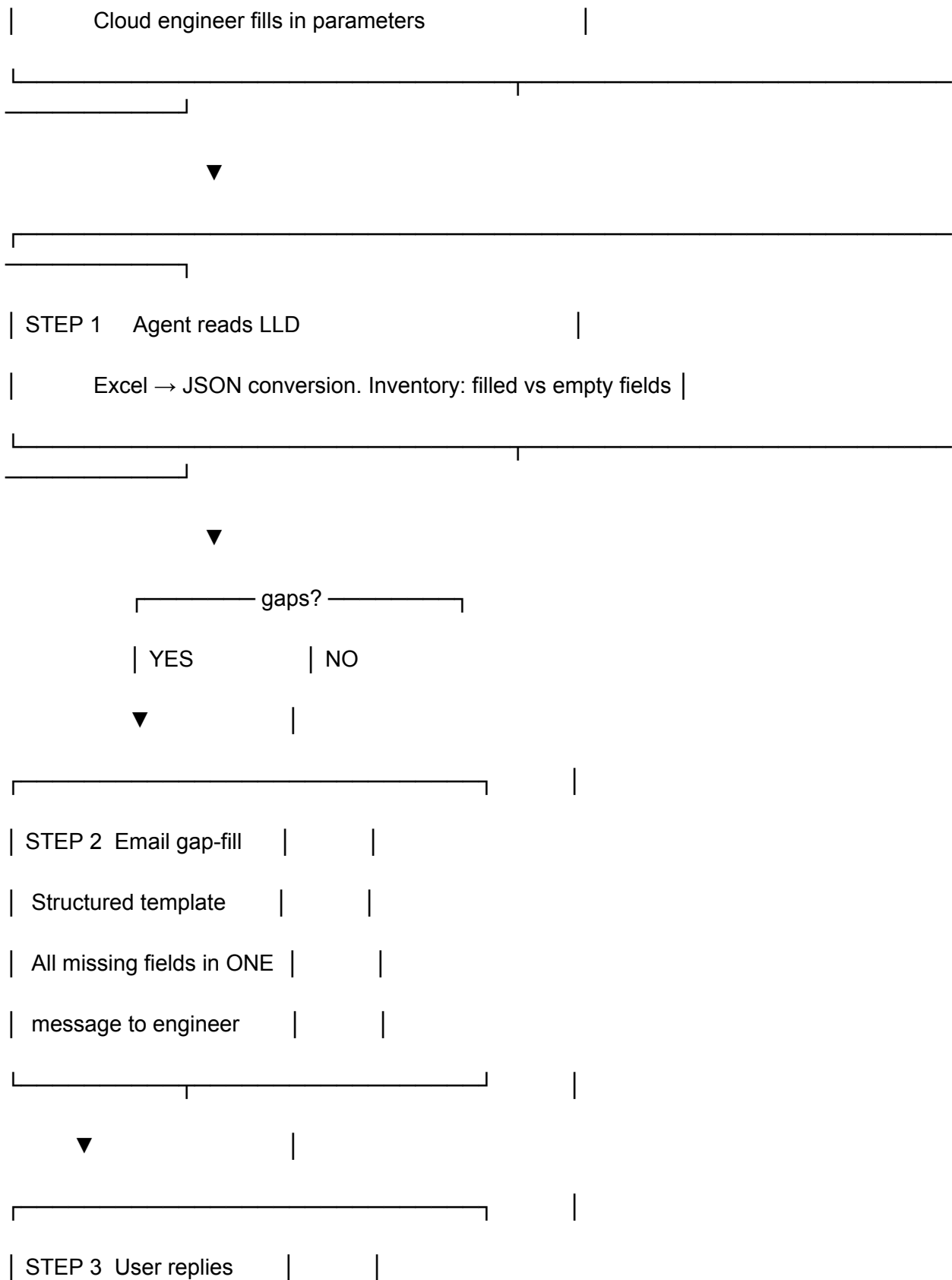
5. Users and Personas

Persona	Description	Interaction with system
Cloud Engineer	Internal infrastructure engineer, comfortable with cloud concepts, may not know Huawei specifics	Fills Excel LLD, replies to gap-fill email, clicks APPROVE in approval email
PM-Lead / Approver	Owens the deployment decision	Receives and approves the plan email. May be same person as engineer in small teams
Security / Compliance	Reviews the LLD before submission	Signs off on the filled Excel LLD before it enters the agent workflow
Audit / Operations	Reviews CTS logs and conformance reports post-deployment	Reads the completion report, accesses CTS audit trail

6. Complete Workflow

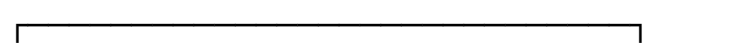
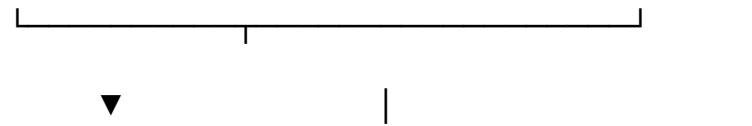
Overview diagram





Fills field_name=[value]

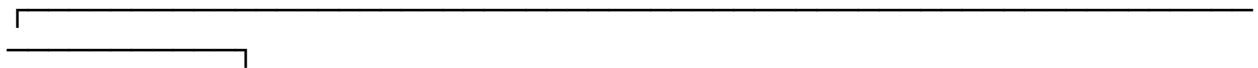
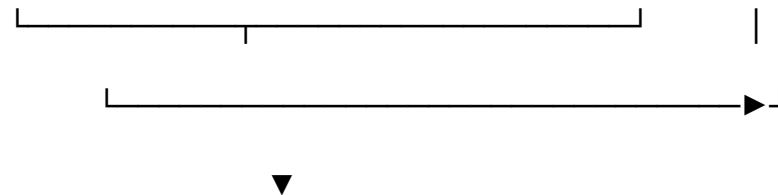
| brackets and replies |



| STEP 4 Agent parses reply |

Regex → update JSON LLD

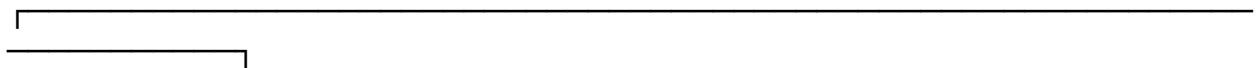
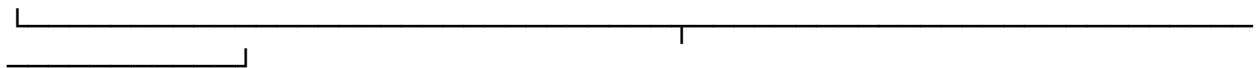
```
| Loop if still gaps (max 2) |
```



STEP 5 Technical Validation (OPA + JSON Schema)

Hard blockers: region, CIDR, required fields, SCP v5

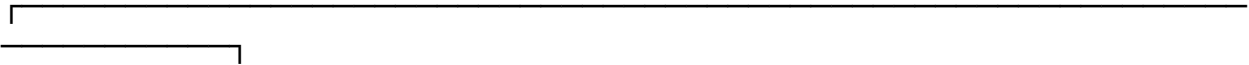
FAIL → short correction email to engineer



STEP 6 CAF Advisory Check

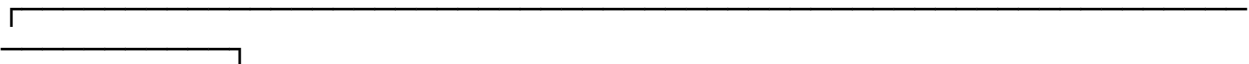
Warnings: SecMaster, log retention, Identity Center

Not blockers — approver acknowledges or defers with reason



STEP 7 Generate Artifacts (deterministic — no AI)

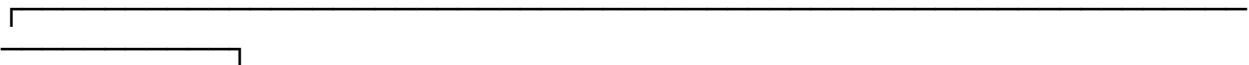
JSON → terraform.tfvars + lld-summary.md + lld-validated.yaml |



STEP 8 Terraform Plan

```
terraform plan -out=tfplan
```

Agent reads output → plain-English summary + Infracost



STEP 9 Approval Email

Plan summary + resource count + cost delta → engineer

Engineer replies: APPROVE or REJECT: reason

| APPROVE



| STEP 10 CI/CD Executes Apply

| GitHub Actions / CodeArts Pipeline

| OIDC token → Huawei IAM STS → 30-min AK/SK

| terraform apply → Huawei Cloud REST APIs

| Agent never holds credentials



| STEP 11 Post-Apply Verification

| Agent queries Huawei APIs:

| RGC status=ENABLED, CTS tracker active, ER attached,

| Conformance Pack deployed and reporting



DONE	Completion Report Email	
	Run ID, wall-clock time, CTS audit trail link	
	Landing Zone is live	

Step-by-step specification

INPUT — Excel LLD file

The cloud engineer (or the pre-sales / cloud architecture team) fills in a structured Excel workbook. One sheet per governance domain, one row per Terraform variable. Columns: `field_name`, `value`, `required`, `default`, `hint`, `validation_rule`.

The Excel is the **human-facing contract**. It is reviewed and signed off by the Security/Compliance team before entering the agent workflow. Once submitted, it is immutable — changes require a new run-id.

STEP 1 — Agent reads LLD

The agent (Claude Code session, running in ephemeral Docker) receives the Excel file path. It:

1. Converts Excel → JSON using `openpyxl` (Python) or `sheetjs` (Node)
2. Validates JSON structure against the master JSON Schema
3. Runs an inventory: which required fields are empty, which have defaults, which are filled
4. Checks cross-field consistency (e.g. region vs CTS global service constraint)
5. Produces: `{filled: [...], empty: [...], inconsistencies: [...]}`

If `empty.length === 0` and `inconsistencies.length === 0`, skip Steps 2–4 and go directly to Step 5.

STEP 2 — Email gap-fill (Gmail channel, MVP)

The agent sends one structured email to the LLD owner (email from Excel metadata field). The email contains:

- A module-grouped list of all missing/ambiguous fields
- Each field on its own line: `field_name = [ ] # hint`
- Fields with defaults pre-filled in green: `field_name = [ default_value ] # CAF default – change if needed`

- CAF advisory warnings listed at the bottom
- Reply rules: fill brackets, reply to thread, defer advisories with `# key: deferred - reason`

Critical rule: All missing fields in ONE email. Never drip one question at a time.

From address: `lz-agent@yourcompany.com` (dedicated mailbox) Reply-to: same address (agent monitors this inbox) Subject format: `[LZ-{run_id}] Missing fields - fill and reply to this email`

STEP 3 — User fills and replies

User replies to the same email thread. They fill in `[ value ]` brackets. Comments after `#` are ignored. They can defer CAF advisories with `# advisory_name: deferred - reason`. No other format changes allowed.

STEP 4 — Agent parses reply

Agent polls the dedicated inbox (every 5 minutes, 24-hour timeout) for a reply to this run's thread. On receipt:

1. Strips email client quoting (lines starting with `>`)
2. Applies regex: `^(\\w+)\\s*=\\s*\\[\\[\\^\\]\\]*\\]` to extract field=value pairs
3. Applies regex: `^#\\s*(\\w+):\\s*deferred\\s*-\\s*(.+)\\$` to extract deferrals
4. Merges values into JSON LLD
5. Re-checks for still-empty required fields
6. If still-empty > 0: sends a short follow-up email (only those fields) — loops back
7. If all filled: proceeds to Step 5

Maximum 2 loop iterations. If Round 3 is needed, flag as "LLD template needs improvement" and escalate.

STEP 5 — Technical validation (OPA + JSON Schema)

Hard blockers — run BEFORE any Terraform is generated. All must pass before proceeding.

Check	Tool	Fail action
Required fields populated	JSON Schema	Correction email
CIDR format valid	OPA Rego	Correction email
Region in allowlist	OPA Rego	Correction email
Email format valid	JSON Schema	Correction email

Check	Tool	Fail action
SCP content uses v5 syntax	OPA Rego	Correction email
No conflicting CIDR ranges	OPA Rego	Correction email
Log archive email $\neq$ audit email	OPA Rego	Correction email

Correction email: same structured format as gap-fill, but only shows the fields with errors. Annotated with the specific validation error message.

STEP 6 — CAF advisory check

Non-blocking warnings that the approver must acknowledge. Checked against a Huawei CAF compliance checklist:

Advisory	Condition	Severity
SecMaster not enabled	<code>enable_secmaster = false</code>	WARN
Log retention < 3 years	For regulated industries	WARN
Identity Center not enabled	<code>enable_identity_center = false</code>	WARN
HSS quotas not set	<code>hss_quotas = 0</code>	WARN
Conformance pack not L1	Not using L1 Security pack	INFO

All unacknowledged advisories are shown in the approval email (Step 9). The approver can override each with a documented reason. Deferrals are recorded in the audit trail.

STEP 7 — Generate artifacts (deterministic, no AI)

A deterministic Python/Node script converts the validated JSON LLD into:

1. `terraform.tfvars` — one file per environment composition (01-foundation, 02-network, etc.)
2. `lld-summary.md` — human-readable LLD summary for sign-off and audit
3. `lld-validated.yaml` — machine-readable archive committed to Git

This step uses no AI. Output must be 100% reproducible given the same JSON input.

STEP 8 — Terraform plan

Agent runs `terraform plan -out=tfplan` against the generated tfvars in a sandboxed environment. Also runs the validation gate sequence:

1. `terraform fmt -check`
2. `terraform validate`
3. `tflint`
4. `checkov -d .`
5. `conftest test --policy ./policies/opa .`
6. `terraform plan -out=tfplan`
7. `infracost breakdown --path=tfplan`

If any gate fails, agent self-corrects up to 3 times (fixes the generated tfvars or module composition), then escalates via email if still failing.

Agent reads the plan output and writes a plain-English summary: how many resources to create/update/destroy, estimated monthly cost delta, and any resources the approver should specifically review.

STEP 9 — Approval email

Agent sends approval email to the LLD owner (and optionally a second approver):

Subject: [LZ-{run_id}] Plan ready — APPROVE or REJECT

PLAN SUMMARY

+ 23 resources to CREATE

~ 0 resources to UPDATE

X 0 resources to DESTROY ← always verify this is 0

ESTIMATED COST

Monthly delta: +\$48/month

KEY RESOURCES

+ huaweicloud_rgc_landing_zone.this

+ huaweicloud_er_instance.hub

[... 5 more shown, full plan at link]

CAF ADVISORIES DEFERRED

secmaster: deferred — will enable in phase 2

Reply with: **APPROVE** to proceed

REJECT: <reason> to cancel

Agent polls for reply. On **APPROVE**: release CI/CD trigger. On **REJECT**: **reason**: cancel run, log reason, send cancellation confirmation.

Approver identity is captured from the email From header and written to the run audit record.

This gate is non-negotiable — never auto-approve in production.

STEP 10 — CI/CD executes apply

Approval triggers the CI/CD pipeline (GitHub Actions or Huawei CodeArts Pipeline):

1. Checkout infra repo (sparse: only workspace/{run_id}/)
2. Set up Terraform (pinned version from **.terraform-version**)
3. Exchange OIDC token with Huawei IAM STS → temporary AK/SK (30-min TTL)
4. **terraform init** (OBS S3-compatible backend)
5. **Pre-apply guard**: fail if any **resource "huaweicloud_*** block found outside a module
6. **terraform state pull > backup-{run_id}.tfstate** → upload to OBS backup prefix
7. **terraform apply tfplan** (pre-approved locked plan)

The agent **never holds credentials**. Credentials exist only inside the CI/CD runner for the duration of the apply step.

STEP 11 — Post-apply verification

Agent queries Huawei Cloud APIs directly to verify key resources are live:

Check	API query	Expected
RGC landing zone status	GET /v1/landing-zone/{id}	status = ENABLED

Check	API query	Expected
CTS org tracker active	GET /v3/{project}/tracker	status = enabled
Enterprise Router exists	GET /v3/{project}/enterprise-router/instances	≥ 1 instance
ER has VPC attachment	GET /v3/{project}/enterprise-router/{id}/vpc-attachments	≥ 1 attachment
Conformance Pack deployed	GET /v1/resource-manager/packages	L1 pack present

If any check fails: send alert email immediately. Do not silently pass. Partial failures must be surfaced.

DONE — Completion report

Agent sends completion email:

Landing Zone deployment complete.

Run ID: LZ-20260524-001

Duration: 27 minutes 14 seconds

Region: cn-east-3

Resources: 23 created, 0 modified, 0 destroyed

CTS audit trail: [https://console.huaweicloud.com/cts/...](https://console.huaweicloud.com/cts/)

State file: obs://lz-tfstate/envs/01-foundation/terraform.tfstate

LLD archive: obs://lz-tfstate/audit/LZ-20260524-001/lld-validated.yaml

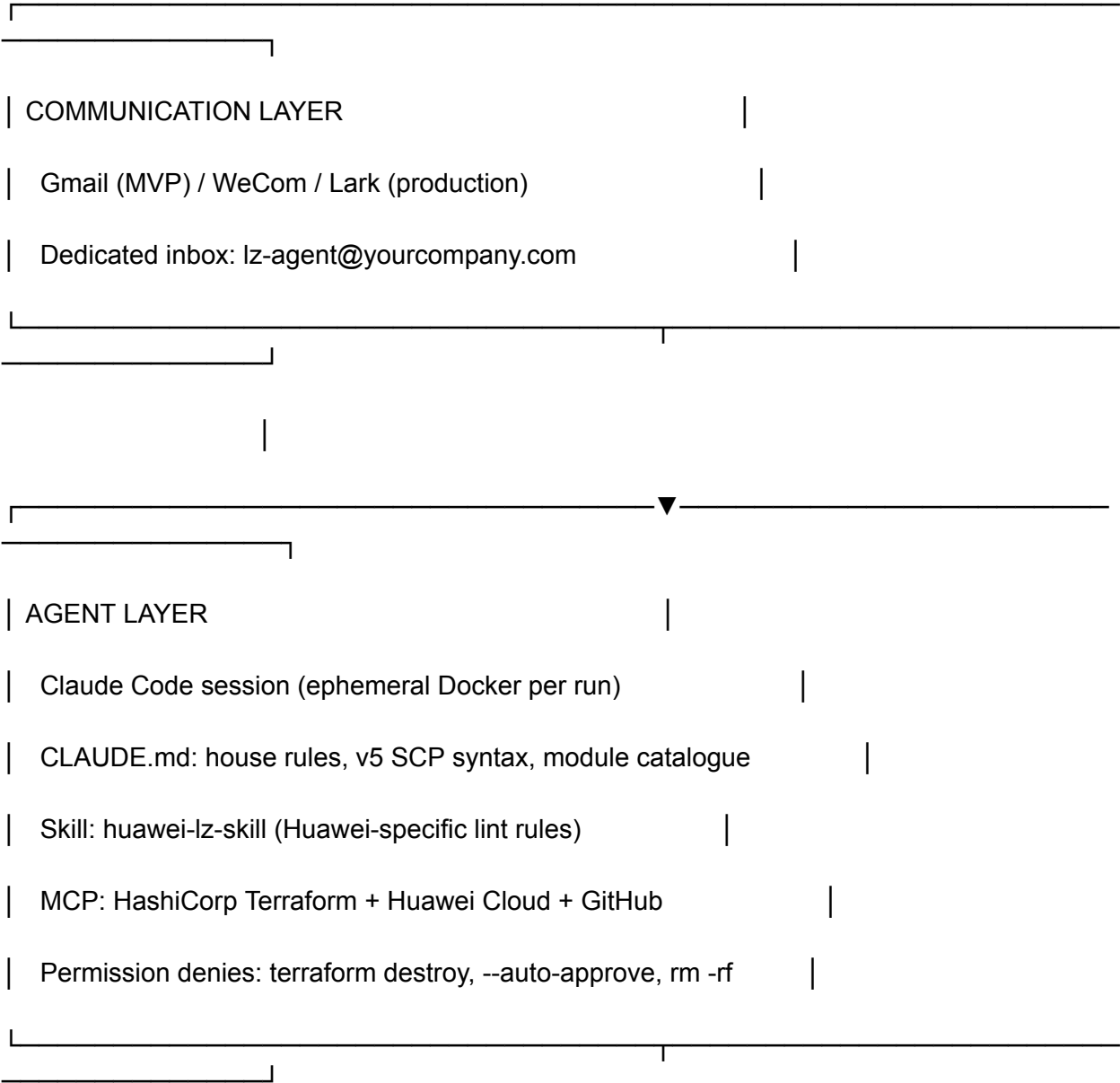
Next steps:

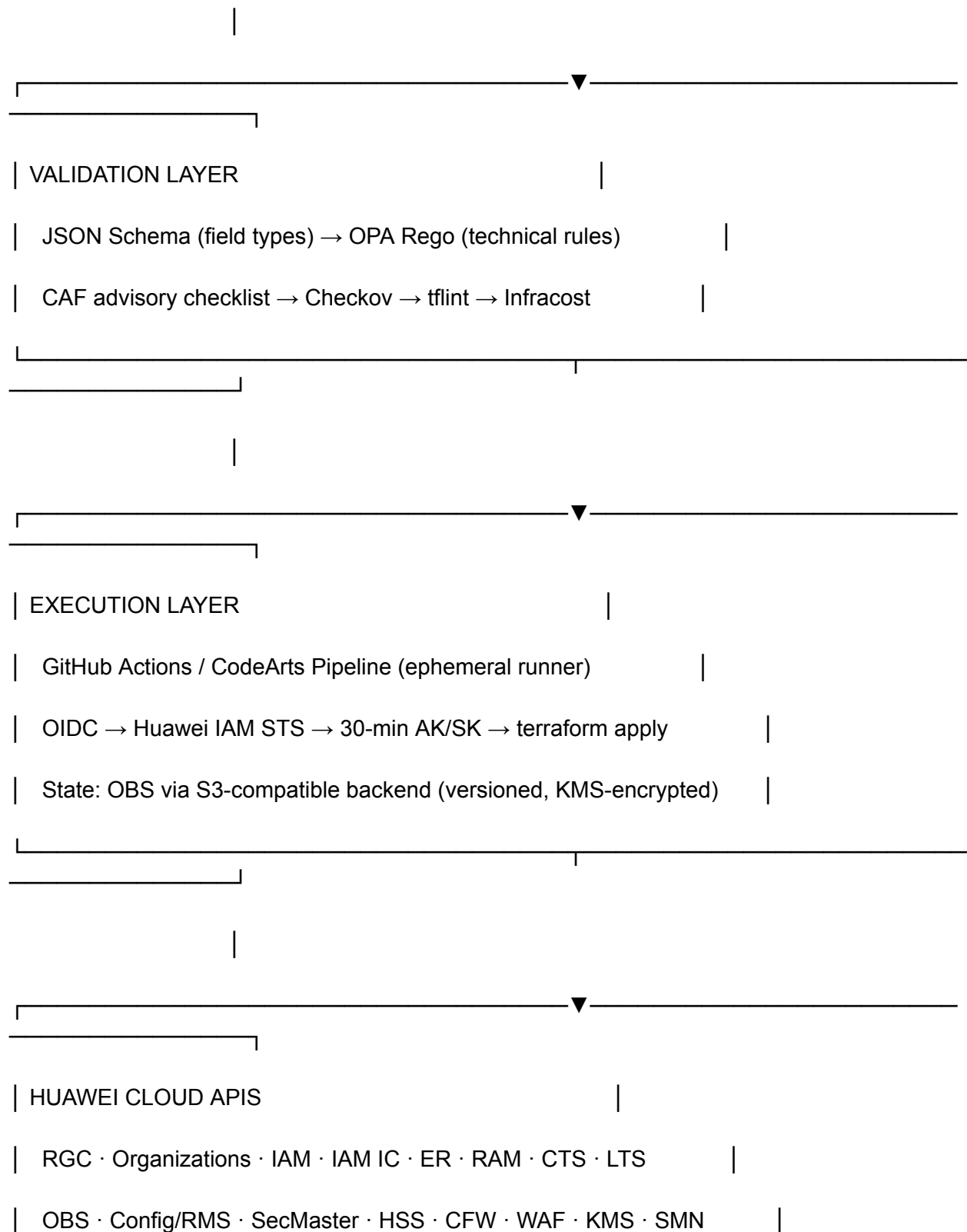
1. Enroll workload accounts via RGC Account Factory

- 2. Assign IAM Identity Center users to permission sets
- 3. Enable SecMaster (deferred from this run)

7. Architecture Overview

System components





Data flow

Excel LLD → JSON (master format) → terraform.tfvars

→ lld-summary.md (human review)

→ lld-validated.yaml (git archive)

→ tfplan (locked plan)

Key design decisions

Decision	Rationale
Excel as input format	Familiar to cloud engineers and compliance teams; convertible to JSON
JSON as master internal format	One parser, multiple outputs; validatable with JSON Schema
Email (Gmail MVP) as communication channel	Zero infrastructure setup; familiar to users; thread-trackable by run-id
Batch all questions in one email	UX: drip questions kill adoption
Separate technical validation from CAF advisory	Technical = hard fail; CAF = acknowledged warning
Deterministic script for tfvars generation	This step must be 100% reproducible — no AI
Approval gate before every apply	Non-negotiable safety checkpoint
CI/CD runner holds credentials, not agent	Agent compromise cannot access cloud credentials
Module-only architecture	Agent composes modules; never writes resource blocks directly
OBS S3-compatible backend	No native HWC TF backend; requires 5 skip_* flags

Decision	Rationale
CTS org tracker hard-coded to cn-north-4	Global service constraint

8. Huawei Cloud Service Inventory

All 9 CAF governance domains, 45+ services, each mapped to Terraform resources.

Domain 1 — Organization & account management

Service	Terraform resources
Organizations	<code>huaweicloud_organizations_organization</code> , <code>_organizational_unit</code> , <code>_account</code> , <code>_policy</code> , <code>_policy_attach</code> , <code>_trusted_service</code>
Resource Governance Center (RGC)	<code>huaweicloud_rgc_landing_zone</code> (full CRUD v1.87+), <code>_operation</code> , <code>_pre_launch_check</code>
Enterprise Center	<code>huaweicloud_enterprise_project</code>

Domain 2 — Identity & permissions

Service	Terraform resources
IAM	<code>huaweicloud_identity_user</code> , <code>_group</code> , <code>_group_membership</code> , <code>_policy</code> , <code>_role</code> , <code>_agency</code> , <code>_provider</code> , <code>_password_policy</code> , <code>_login_policy</code> , <code>_acl</code>
IAM Identity Center	<code>huaweicloud_identitycenter_user</code> , <code>_group</code> , <code>_permission_set</code> , <code>_account_assignment</code> , <code>_provisioned_permission_set</code>

Domain 3 — Centralized network

Service	Terraform resources
VPC	<code>huaweicloud_vpc</code> , <code>_vpc_subnet</code> , <code>_vpc_route</code> , <code>_vpc_peering_connection</code> , <code>_vpc_flow_log</code>
Security Groups / ACL	<code>huaweicloud_networking_secgroup</code> , <code>_secgroup_rule</code> , <code>_network_acl</code> , <code>_network_acl_rule</code>
Enterprise Router (ER)	<code>huaweicloud_er_instance</code> , <code>_vpc_attachment</code> , <code>_route_table</code> , <code>_static_route</code> , <code>_propagation</code> , <code>_association</code> , <code>_flow_log</code>
Cloud Connect (CC)	<code>huaweicloud_cc_central_network</code> , <code>_bandwidth_package</code> , <code>_connection</code>
Direct Connect (DC)	<code>huaweicloud_dc_connection</code> , <code>_virtual_gateway</code> , <code>_virtual_interface</code>
VPN	<code>huaweicloud_vpn_gateway</code> , <code>_connection</code> , <code>_customer_gateway</code>
NAT	<code>huaweicloud_nat_gateway</code> , <code>_snat_rule</code> , <code>_dnat_rule</code>
EIP	<code>huaweicloud_vpc_eip</code> , <code>_vpc_bandwidth</code> , <code>_vpc_eip_associate</code>
ELB	<code>huaweicloud_elb_loadbalancer</code> , <code>_listener</code> , <code>_pool</code> , <code>_member</code> , <code>_certificate</code>
DNS	<code>huaweicloud_dns_zone</code> , <code>_recordset</code> , <code>_endpoint</code> , <code>_resolver_rule</code>

Domain 4 — Resource sharing

Service	Terraform resources
RAM	<code>huaweicloud_ram_resource_share,</code> <code>_resource_share_accepter</code>
OBS (shared)	<code>huaweicloud_obs_bucket,</code> <code>_bucket_policy,</code> <code>_bucket_lifecycle_rule,</code> <code>_bucket_object</code>
SFS Turbo	<code>huaweicloud_sfs_turbo</code>
IMS	<code>huaweicloud_images_image,</code> <code>_images_image_share</code>
CCM	<code>huaweicloud_ccm_certificate,</code> <code>_ccm_private_ca,</code> <code>_ccm_private_certificate</code>

Domain 5 — Unified security

Service	Terraform resources
SecMaster	<code>huaweicloud_secmaster_workspace,</code> <code>_alert_rule,</code> <code>_playbook,</code> <code>_playbook_action</code>
HSS	<code>huaweicloud_hss_host_group,</code> <code>_host_protection,</code> <code>_quotas</code>
Cloud Firewall (CFW)	<code>huaweicloud_cfw_firewall,</code> <code>_protection_rule,</code> <code>_address_group,</code> <code>_acl_rule,</code> <code>_eip_protection</code>
WAF	<code>huaweicloud_waf_dedicated_instanc</code> <code>e,</code> <code>_dedicated_domain,</code> <code>_policy,</code> <code>_rule_*</code>
DEW / KMS	<code>huaweicloud_kms_key,</code> <code>_kms_key_grant,</code> <code>huaweicloud_csms_secret</code>

Service	Terraform resources
DBSS	<code>huaweicloud_dbss_instance</code> , <code>_audit_database</code>
DSC	<code>huaweicloud_dsc_instance</code> (partial coverage)

Domain 6 — Unified compliance audit

Service	Terraform resources
CTS	<code>huaweicloud_cts_tracker</code> (<code>is_organization_tracker=true</code> , must be in cn-north-4), <code>_data_tracker</code> , <code>_notification</code>
LTS	<code>huaweicloud_lts_group</code> , <code>_stream</code> , <code>_transfer</code> , <code>_structuring_configuration</code> , <code>_sql_alarm_rule</code> , <code>_keywords_alarm_rule</code>
OBS (audit archive)	<code>huaweicloud_obs_bucket</code> , <code>_bucket_lifecycle_rule</code>
Config / RMS	<code>huaweicloud_rms_policy_assignment</code> , <code>_assignment_package</code> , <code>_organizational_policy_assignment</code> , <code>_resource_aggregator</code>

Domain 7 — Unified O&M

Service	Terraform resources
Cloud Eye (CES)	<code>huaweicloud_ces_alarmrule</code> , <code>_resource_group</code> , <code>_alarm_template</code> , <code>_dashboard</code>
AOM	<code>huaweicloud_aom_alarm_rule</code> , <code>_service_discovery_rule</code> , <code>_prometheus_instance</code>

Service	Terraform resources
SMN	<code>huaweicloud_smn_topic,</code> <code>_subscription, _topic_policy</code>
FunctionGraph	<code>huaweicloud_fgs_function, _trigger,</code> <code>_async_invoke_configuration</code>

Domain 8 — Financial management

Service	Terraform resources
Tag policies	<code>huaweicloud_organizations_policy</code> (<code>type = "tag_policy"</code>)
BSS / Cost Center	Data sources only (console-managed for MVP)

Domain 9 — Data perimeters

Service	Terraform resources
SCPs	<code>huaweicloud_organizations_policy</code> (<code>type = "service_control_policy"</code> , Version 5.0)
VPC Endpoints (VPCEP)	<code>huaweicloud_vpcep_service,</code> <code>_endpoint, _approval</code>
OBS bucket policies	<code>huaweicloud_obs_bucket_policy</code> with Condition blocks

9. Terraform Module Specifications

Module 1: `01-org-foundation`

Purpose: RGC bootstrap + Organizations extensions

Key resource: `huaweicloud_rgc_landing_zone` (full CRUD, v1.87+)

Deploy time: ~25 minutes (RGC bootstrap)

Outputs: `root_id`, `core_ou_id`, `log_archive_account_id`, `audit_account_id`,

`management_account_id`

Critical constraint: RGC must be enabled FIRST. All other modules depend on its outputs.

Module 2: `02-identity-center`

Purpose: IAM Identity Center, permission sets, account assignments

Standard permission sets shipped: `LzAdministrator`, `LzDeveloper`,
`LzSecurityAuditor`, `LzBillingViewer`, `LzReadOnly`

Pattern: `for_each` over a `permission_sets` map variable

Module 3: `03-iam-baseline`

Purpose: Per-account IAM hardening — password policy, login policy, root user protection, baseline agencies

Note: Uses `count` to apply across all governed accounts via aliased providers

Module 4: `04-network-hub`

Purpose: Enterprise Router (hub), Cloud Firewall, NAT, shared bandwidth, RAM cross-account share

Provider: `huaweicloud.network_ops` (alias)

Key resource: `huaweicloud_er_instance` + `huaweicloud_ram_resource_share`
(shares ER to all spoke accounts)

Dependency: RAM share must be accepted by spoke accounts before VPC attachment
(handled by `_resource_share_accepter`)

Module 5: `05-network-spoke`

Purpose: Per-account VPC + ER attachment

Pattern: Called once per spoke account with a different provider alias

Key resources: `huaweicloud_vpc`, `huaweicloud_er_vpc_attachment`

Modules 6–13 (abbreviated)

Module	Key services	Notes
<code>06-public-services</code>	ELB, WAF, DNS, CCM	WAF sits in <code>network_ops</code> account
<code>07-shared-resources</code>	KMS, OBS shared, SFS, IMS	KMS keys shared via RAM
<code>08-security-center</code>	SecMaster, HSS, DBSS, CSMS	HSS quotas in <code>security_ops</code> , agents per-account

Module	Key services	Notes
09-audit-logging	CTS org tracker, LTS, OBS archive	CTS MUST be in cn-north-4
10-compliance-config	Config/RMS, conformance packs	L1 Security Conformance Pack by default
11-ops-monitoring	Cloud Eye, AOM, SMN, FunctionGraph	Alerting routed to SMN topics
12-finance-governance	Tag policies, BSS data sources	Tag policies attached at org root
13-data-perimeter	SCPs, VPCEP, OBS bucket policies	<code>enforce_mode=false</code> until VPCEPs deployed

10. Email Template Specification

Gap-fill email format

Subject: [LZ-`{run_id}`] Missing fields — fill and reply to this email

From: lz-agent@yourcompany.com

Reply-To: lz-agent@yourcompany.com

MODULE: `{module_name}`

`{field_name:<35}` = [] # REQUIRED — `{hint}`

`{field_name:<35}` = [`{default}`] # CAF default — change if needed

CAF ADVISORY WARNINGS

⚠ `{advisory}`: `{message}`

To defer: # `{advisory}`: deferred — `<reason>`

REPLY RULES

- Reply to THIS thread
- Fill values inside [] brackets only
- Comments after # are ignored
- To defer: # advisory_name: deferred — <reason>

Approval email format

Subject: [LZ-{run_id}] Plan ready — APPROVE or REJECT

PLAN SUMMARY

+ {create_count} resources to CREATE

~ {update_count} resources to UPDATE

X {destroy_count} resources to DESTROY

ESTIMATED COST: {cost_delta}/month

Reply: APPROVE or REJECT: <reason>

Parsing rules

Pattern	Action
<code>^(\w+)\s*=\s*\[([^\]]*)\]</code>	Extract field=value pair
<code>^>.*\$</code>	Strip (email client quoting)
<code>^#\s*(\w+):\s*deferred\s*-\s*(.+) \$</code>	Record deferral
Empty brackets []	Add to still_missing list
Unknown field name	Log warning, skip silently

11. Security Model

Credential flow

GitHub OIDC token

- Huawei IAM STS (AssumeRoleWithWebIdentity)
- LzAgentDeployAgency (30-min AK/SK)
- terraform apply
- per-account OrganizationAccountAccessAgency
- Huawei Cloud APIs

No static credentials stored anywhere. Agent never holds credentials. Credentials expire automatically.

Threat mitigations

Threat	Mitigation
Malicious LLD content	OPA + JSON Schema validation; normalizer is separate from executor
Agent runs terraform destroy	Hard permission deny in Claude Code config (<code>Bash(terraform destroy*)</code>)
State corruption from concurrent runs	CI/CD concurrency groups serialize per workspace
Long-lived credentials leaked	OIDC keyless; 30-min TTL; no static AK/SK in CI secrets
Unauthorized email approval	Approver list checked against allowlist; identity logged to CTS
Inline resource block outside module	Pre-apply guard shell script fails the run

12. Development Roadmap

Phase 1 — Terraform Library (Weeks 1–8)

Week	Deliverable
1	OBS state backend works, OIDC auth to all accounts works, provider v1.87 confirmed
2	Modules 01–03 complete and tested in sandbox
3	Modules 04–07 complete and tested
4–5	Modules 08–10 complete and tested
6	Modules 11–13 complete and tested
7	Full end-to-end apply in sandbox (all 13 modules, 5 env compositions)
8	CI/CD pipeline complete, all gates passing, acceptance tests done

Phase 2 — Agentic Workflow (Weeks 9–14)

Week	Deliverable
9	Excel LLD template complete (all 13 modules, all variables documented)
10	JSON Schema + OPA policy library complete; Excel → JSON converter
11	Agent: LLD parsing, gap detection, email send/receive/parse loop
12	Agent: tfvars generation, terraform plan, approval email flow
13	CI/CD trigger from approval, post-apply verification, completion email

Week	Deliverable
14	End-to-end test: Excel LLD → deployed LZ, < 30 min, 0 manual steps

Phase 3 — Production Hardening (Weeks 15–18)

- WeCom/Lark channel replaces Gmail
- Multi-run concurrency testing
- Drift detection agent
- Account vending automation
- Chaos tests: state corruption, mid-apply kill, approval timeout

13. Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
Huawei RGC bootstrap takes > 25 min	Medium	Low	Accepted; idempotent on retry
OBS state corruption (no native locking)	Medium	High	CI concurrency groups + OBS versioning + pre-apply backup
SCP v5 syntax errors	High (W1)	Medium	CLAUDE.md cheat sheet + OPA rule
CTS org tracker region constraint missed	Certain	Low	Hard-coded in module, not a variable
Agent generates resource outside module	Low (with guard)	Medium	Pre-apply shell check fails the run
Scope creep mid-build	High	High	Feature freeze EOW6 (Terraform) and EOW12 (Agent)

Risk	Likelihood	Impact	Mitigation
Email reply not parsed correctly	Medium	Low	Structured template + regex parsing + round-2 follow-up
User ignores emails / slow reply	Medium	Medium	24-hour timeout with escalation email to PM-lead
Provider < v1.87 missing RGC resource	Certain if not pinned	High	<code>version = "~>1.87"</code> in all modules

14. Team Operating Model

Roles

Role	Responsibility
PM-Lead	Scope, approvals, success-metric scoreboard, case-study writeup
Engineer A (Platform/Terraform)	Modules 01–13, OBS state backend, provider correctness
Engineer B (Agent/Email)	Claude Code harness, email reader/sender, OPA policies
Engineer C (SecOps/CI)	OIDC + IAM agencies, GitHub Actions workflow, observability

If only 2 engineers: A absorbs CI, B absorbs SecOps.

Agent-first habits

1. **Ask the agent first** — even if you reject the output. The default question changes from "how do I write this?" to "what does the agent need to know to write it?"
2. **PRs include agent transcripts** — reviewers must see the prompt-context that produced the change

3. **Failures improve CLAUDE.md, not patches** — every failure improves the team's collective prompt

Metrics to track

Metric	Target trend
Cycle time per deployment	Falling
Agent self-correction rate	Rising
Reviewer touch-time on agent output	Falling
Defect escape rate (agent vs manual)	Equal or lower

Weekly rhythm

- Daily 15-min standup (scope, blockers)
 - Wednesday: mid-week demo to one external stakeholder
 - Friday: 30-min prompt retro — pick worst failure, update CLAUDE.md
-

15. Acceptance Criteria

The product is "done" when all the following hold across **three consecutive runs** on fresh Huawei accounts:

1. A filled Excel LLD → fully deployed baseline LZ with zero console steps
 2. Wall-clock time < 30 minutes
 3. CTS audit shows zero out-of-band human actions in the run window
 4. All OPA gates pass; at least one blocked a deliberately-broken LLD variant
 5. State is recoverable from OBS versioning in < 5 minutes
 6. Full audit trail: CTS + agent transcript + approved plan archived in OBS
 7. Completion email delivered with correct run-id, timing, and CTS link
-

16. Glossary

Term	Definition
LLD	Low-Level Design — the structured Excel document that drives a deployment
RGC	Resource Governance Center — Huawei's Control Tower equivalent
OU	Organizational Unit — a grouping within Huawei Cloud Organizations
SCP	Service Control Policy — preventive guardrail applied to an OU
ER	Enterprise Router — Huawei's Transit Gateway equivalent
CTS	Cloud Trace Service — Huawei's CloudTrail equivalent
LTS	Log Tank Service — Huawei's CloudWatch Logs equivalent
OBS	Object Storage Service — Huawei's S3 equivalent
CAF	Cloud Adoption Framework — Huawei's reference architecture for enterprise cloud adoption
OIDC	OpenID Connect — used for keyless CI/CD credential exchange with Huawei IAM
OPA	Open Policy Agent — policy-as-code engine used for Terraform plan validation
tfvars	Terraform variable values file — generated from LLD JSON